Experiment No:	2
Subject:	Database Management Laboratory
Title:	Design and develop at SQL DDL statements which demonstrate the use of SQL objects such as Table, View, Index, Sequence and Synonym.
Roll No.:	
Class:	TE (E&TC)
Division:	
Signature:	

PROGRESSIVE EDUCATION SOCIETY
Pune - 5
Modern College of Engineering
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Group A- Database Programming Languages – SQL

Experiment No. -2

Title: Design and develop at least 5 SQL DDL statements which demonstrate the use of SQL objects such as Table, View, Index, Sequence and Synonym.

Objective: To learn all type Data Definition Language commands and their uses.

Hardware requirements: Any CPU with Pentium Processor or similar, 256 MB RAM or more, 1 GB Hard Disk or more.

Software requirements: Windows Operating System, MySQL

Theory:

Introduction to SQL:

- SQL stands for Structured Query Language
- SQL lets you access and manipulate databases
- SQL is an ANSI (American National Standards Institute) standard

Commands of SQL are grouped into four languages.

DDL:

DDL is abbreviation of Data Definition Language. It is used to create and modify the structure of database objects in database.

Examples: CREATE, ALTER, DROP, RENAME, TRUNCATE statements

DML

DML is abbreviation of Data Manipulation Language. It is used to retrieve, store, modify, delete, insert and update data in database.

Examples: SELECT, UPDATE, INSERT, DELETE statements

DCL

DCL is abbreviation of Data Control Language. It is used to create roles, permissions, and referential integrity as well it is used to control access to database by securing it.

Examples: GRANT, REVOKE statements

TCL

TCL is abbreviation of Transactional Control Language. It is used to manage different transactions occurring within a database.

Examples: COMMIT, ROLLBACK statements

Data Definition Language (DDL)

1.Data definition Language (DDL) is used to create, rename, alter, modify, drop, replace, and delete tables, Indexes, Views,

and comment on database objects; and establish a default database.

2.The DDL part of SQL permits database tables to be created or deleted. It also defines indexes (keys), specify links

between tables and impose constraints between tables. The most important DDL statements in SQL are:

- CREATE TABLE- Creates a new table
- ALTER TABLE- Modifies a table
- DROP TABLE- Deletes a table
- TRUNCATE -Use to truncate (delete all rows) a table.
- CREATE INDEX- Creates an index (search key)
- DROP INDEX- Deletes an index

1.**The CREATE TABLE Statement**

The CREATE TABLE statement is used to create a table in a database.

Syntax

CREATE TABLE tablename

(attr1_name attr1_datatype(size) attr1_constraint,

attr2_name attr2_datatype(size) attr2_constraint,….);

SQL Constraints

Constraints are used to limit the type of data that can go into a table.

Constraints can be specified when a table is created (with the CREATE TABLE statement) or after the table is created (with the ALTER TABLE statement).

We will focus on the following constraints:

NOT NULL

UNIQUE

PRIMARY KEY

FOREIGN KEY

CHECK

DEFAULT

Add constraint after table creation using alter table option

Syntax - Alter table add constraint constraint_name constraint_type (Attr_name) Example - Alter table stud add constraint

prk1 primary key(rollno);

Drop constraint:

Syntax- Drop Constraint Constraint_name;

Example - Drop constraint prk1;

2.The Drop TABLE Statement

Removes the table from the database

Syntax

DROP TABLE table_name;

3. **The ALTER TABLE Statement**

The ALTER TABLE statement is used to add, delete, or modify columns in an existing table.

Syntax

To add a column in a table, use the following syntax:

ALTER TABLE table_name

ADD column_name datatype;

To delete a column in a table, use the following syntax (notice that some database systems don't allow deleting a column):

ALTER TABLE table_name DROP COLUMN column_name;

To change the data type of a column in a table, use the following syntax:

ALTER TABLE table_name

MODIFY COLUMN column_name datatype;

4. The RENAME TABLE Statement

Rename the old table to new table;

Syntax

Rename old_tabname to new_tabname;

5. The TRUNCATE TABLE Statement

The ALTER TABLE Statement is used to truncate (delete all rows) a table.

Syntax

To truncate a table, use following syntax : TRUNCATE TABLE table_name;

6. CREATE VIEW Statement

In SQL, a view is a virtual table based on the result-set of an SQL statement.A view contains rows and columns, just like a

real table. The fields in a view are fields from one or more real tables in the database.

Syntax

CREATE VIEW view_name AS

SELECT column_name(s)

FROM table_name

WHERE condition;

7. SQL Dropping a View

You can delete a view with the DROP VIEW command.

Syntax

DROP VIEW view_name;

8 . Create Index Statement

1. Index in SQL is created on existing tables to retrieve the rows quickly. When there are thousands of records in a

table, retrieving information will take a long time.

2. Therefore indexes are created on columns which are accessed frequently, so that the information can be

retrieved quickly.

3. Indexes can be created on a single column or a group of columns. When a index is created, it first sorts the

data and then it assigns a ROWID for each row.

Syntax

CREATE INDEX index_name

ON table_name (column_name1,column_name2...);

- index_name is the name of the INDEX.
- table_name is the name of the table to which the indexed column belongs
- column_name1, column_name2.. is the list of columns which make up the INDEX.

9. Drop Index Statement

Syntax

DROP INDEX index_name;

10. Create Synonym statement

1. Use the CREATE SYNONYM statement to create a synonym, which is an alternative name for a table, view,

sequence, procedure, stored function, package, materialized view.

2. Synonyms provide both data independence and location transparency. Synonyms permit applications to

function without modification regardless of which user owns the table or view and regardless of which

database holds the table or view.

3. You can refer to synonyms in the following DML statements: SELECT, INSERT, UPDATE, DELETE

Syntax - Create synonym synonym-name for object-name;

Example-Create synonym synonym_name for table_name

Create synonym t for test

Conclusion:

Experiment No:	3
Subject:	Database Management Laboratory
Title:	Design and develop at least 5SQL queries for suitable database application using SQL DML statements: Insert and Select with operators and functions.
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